Graphic Era
HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Term Evaluation Theory (Even) Semester Regular Examination May 2026

Roll no. 2294038

Name of the Course and semester : B.tech 8th semester
Name of the Paper: Wind and Solar Energy Systems
Paper Code: TOE-805
Time: 1.5Hour

Maximum Marks: 50

Note:
(i) Answer all the questions by choosing any one of the sub-questions
(ii) Each question carries 10 marks.

Q1. (10 Marks)

a. Examine the socio-economic benefits and challenges associated with the large-scale deployment of renewable energy in a developing country like India. CO1

OR

b. Describe various instruments used for measuring solar radiation such as Pyranometer and Pyrheliometer. Explain their working principles CO2

Q2. (10 Marks)

a. Analyze India's renewable energy potential across various sources such as solar, wind, hydro, and biomass. Which sources hold the most promise for the country, and why? **CO1**

OR

b. What is hour angle? How is it calculated? Explain its role in determining solar position. CO2

Q3. (10 Marks)

a. Explain the fundamental need for transitioning to renewable energy sources, linking it directly to the broader concept of sustainable development. CO1

OR

b. With the help of diagrams, explain the relationship between declination angle, hour angle, and solar altitude. CO2

Q4. (10 Marks)

a. Compare and contrast the global, national, and regional energy scenarios, highlighting key trends, challenges, and opportunities in each context. CO1

OR

b. Explain the construction and working of a flat plate collector. Discuss its advantages and limitations.

CO2

Q5. (10 Marks)

a. Discuss the major environmental impacts associated with the continued reliance on fossil fuels, covering aspects such as air quality, climate change, and ecosystem degradation. CO1

OR

b. Describe solar thermal power generation. Explain how electricity is generated using solar thermal energy CO2